A Dynamic Spectrum Sharing and Access Algorithm for Co-Channel Interference Mitigation in 6G Direct-to-Cell (DtC) Networks

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***Abstract*—With the evolution of sixth-generation mobile communication technology (6G) toward an integrated space–air–ground network, Direct-to-Cell (DtC) communication is emerging as a key solution to achieve seamless global coverage, attracting widespread attention from the industry. However, the large-scale deployment of DtC services further exacerbates the already strained spectrum resources. To ensure terminal compatibility and large-scale commercialization, DtC systems must operate within the same or adjacent frequency bands as terrestrial cellular networks, which introduces severe co-channel interference challenges. These include interference between satellite and terrestrial systems, intra-satellite system interference, and inter-satellite system interference. Among them, downlink interference from terrestrial cellular base stations to DtC users is particularly critical due to its high signal strength and broad coverage, becoming a key bottleneck affecting the quality and stability of DtC services. To address this challenge, this paper proposes a dynamic spectrum sharing access algorithm tailored for 6G DtC scenarios. The algorithm leverages fine-grained real-time interference environment data as input for decision-making. It employs an efficient greedy strategy for the allocation of physical resource blocks (PRB) and integrates a constrained water-filling algorithm to optimize transmission power. The simulation results demonstrate that, compared to the baseline schemes based on 3GPP protocols, the proposed algorithm significantly improves the spectrum efficiency of satellite–terrestrial links. This improvement effectively suppresses co-channel interference while improving the communication experience of DtC users, validating the algorithm's potential for future integrated space–air–ground networks.**

***Index Terms*—Co-channel Interference, Direct-to-Cell, Dynamic Spectrum Sharing, Resource Allocation, Spectral Efficiency.**

I. Introduction

ONE of the core visions for sixth-generation (6G) mobile communications is the construction of a globally ubiquitous Space-Terrestrial Integrated Networks (STINs) to meet the demands of a future where everything is intelligently connected [1].As a key enabling technology for this vision, Direct-to-Cell (DtC) communication, based on Low Earth Orbit (LEO) satellite constellations, is rapidly transitioning from a forward-looking concept to a commercial reality. DtC technology allows standard smartphones to establish direct communication links with satellites in remote areas, over oceans, in deserts, or in post-disaster zones where terrestrial cellular coverage is unavailable. This capability offers immense social and commercial value by providing emergency communications, Internet of Things (IoT) connectivity, and essential data services to users worldwide. [2]

However, the large-scale deployment of DtC technology faces a fundamental constraint: the extreme scarcity of spectrum resources. To maximize compatibility with existing ecosystems and minimize the cost of terminal upgrades, emerging DtC services tend to reuse the mature frequency bands already allocated to terrestrial cellular systems (e.g., 4G LTE and 5G NR), such as the S-band or L-band [3]. This spectrum reuse strategy inevitably leads to complex co-channel interference (CCI) scenarios between DtC satellite systems and terrestrial cellular networks. The resulting interference is multi-dimensional and includes:

1) **Interference between satellite and terrestrial systems**—i.e., satellite signals interfering with terrestrial users or terrestrial base station signals interfering with satellite users.

2) **Intra-satellite system interference**, such as interference between different beams or neighboring satellites within the same constellation.

3) **Inter-constellation interference**, which refers to interference among satellite constellations operated by different providers [4].

Among these interference categories, the terrestrial base-station downlink poses a particularly critical threat to DtC downlink reception. Due to the dense deployment of terrestrial base stations and the substantially shorter propagation distances involved, terrestrial downlink signals often exhibit comparatively higher received power at the UE side. These strong terrestrial signals can readily mask or overwhelm the inherently weak satellite downlink received by DtC terminals, leading to severe link-performance degradation and even service interruption. Consequently, terrestrial downlink co-channel interference has become a primary bottleneck constraining the continuity and reliability of DtC services.

Traditional radio resource management (RRM) schemes, such as those designed by 3GPP for terrestrial cellular networks, are fundamentally ill-suited for the highly dynamic DtC environment. These schemes rely heavily on user-reported Channel State Information (CSI) to perform link adaptation and scheduling [5].

The first challenge arises from the aging of feedback information. In DtC communications, a significant propagation delay exists between the satellite and ground users. According to 3GPP TR 38.821 [5], even under typical NTN configurations, the round-trip time (RTT) on the Uu interface already reaches 4–12.9 ms for regenerative payloads and 8–25.8 ms for transparent payloads. In practical DtC deployments, additional satellite–gateway–core-network routing further enlarges the end-to-end delay, making CSI/Channel Quality Indication

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(CQI) information severely outdated by the time it arrives at the satellite scheduler and thus unsuitable for real-time RRM decisions.

Moreover, the high orbital velocity of LEO satellites (approximately 7.5 km/s) introduces rapid channel variations, further exacerbating the limitations of CSI-based feedback mechanisms. As reported in 3GPP TR 38.811 [6], for a LEO satellite at an altitude of 600 km operating around 2 GHz, the maximum Doppler shift can reach ± 48 kHz, with a Doppler drift rate of approximately -544 Hz/s. In addition, the Uu-interface delay drift is around ± 20 μs/s for regenerative payloads and ± 40 μs/s for transparent payloads. Such rapid temporal and frequency variations accelerate the obsolescence of CSI, causing link-adaptation decisions based on delayed feedback to frequently deviate from the true channel state.

The second challenge lies in the coarse granularity of the feedback information. Conventional CQI reports are typically provided at the wideband or subband level and therefore fail to capture the heterogeneous interference conditions across individual physical resource blocks (PRBs). As noted in the technical filing submitted by Starlink to the FCC [7], [8], when wideband CQI is reported, the presence of severely interfered polluted PRBs and comparatively clean PRBs within the same channel band leads to an averaging effect in the reported CQI value. This averaging masks valuable frequency-selective scheduling opportunities, preventing the scheduler from prioritizing high-quality PRBs and may resulting in the allocation of resources to heavily interfered PRBs.

Under such conditions, the conventional CQI reporting mechanism—due to its slow update cycle and coarse granularity—cannot accurately characterize instantaneous PRB-level interference or channel quality. Consequently, legacy 3GPP scheduling strategies are unable to cope with the rapidly time-varying space–terrestrial interference patterns, thus severely limiting the spectrum utilization efficiency of DtC systems.

To address these challenges, this paper proposes a novel dynamic spectrum sharing access algorithm designed to effectively manage and mitigate co-channel interference from terrestrial base stations. Unlike traditional methods that rely on terminal feedback, our algorithm is based on real-time, fine-grained awareness of the interference environment. We assume the availability of an external information source (e.g., an electromagnetic map) that provides accurate, real-time interference power levels for each PRB for each DtC user.

Based on this input, we design a two-stage resource allocation framework:

1) Firstly, a Greedy Strategy based on marginal gain is adopted to allocate PRBs for users, aiming to instantaneously maximize the overall spectral performance of the system.

2) Secondly,the power of the PRB allocated to users is controlled through the constrained water-filling algorithm to maximize the data transmission rate while meeting the total power limit.

The main contribution of this paper is the development of an interference-aware dynamic resource management scheme that requires no terminal feedback and is well-suited to the high-dynamics of DtC environments.

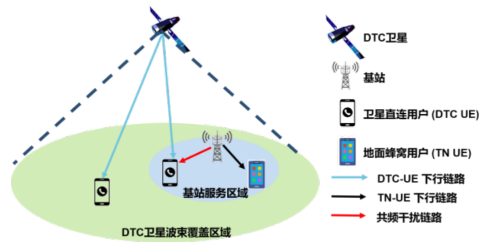

Fig. 1. System Architecture

Simulation results demonstrate that our proposed algorithm achieves significant improvements in spectral efficiency(SE), user throughput, and system fairness compared with a 3GPP-based approaches. Furthermore, the robustness of the algorithm has been investigated under different conditions, including varying radio-map resolutions and user densities.

The remainder of this paper is organized as follows. Section II details the system architecture, channel model, and formulates the resource allocation problem as a Mixed-Integer Non-Linear Program (MINLP). Section III presents our proposed two-stage dynamic spectrum and power allocation algorithm in detail. Section IV describes the simulation methodology, test scenarios, and provides a comprehensive analysis of the performance results. Section V discusses related work in the field of spectrum sharing for STINs. Finally, Section VI concludes the paper and suggests directions for future research.

II. System Model

Within the 3GPP NTN standard framework, this paper investigates a 6G-oriented DtC satellite downlink (DL) communication scenario, focusing on co-channel interference from terrestrial base-station downlink transmissions to DtC users. In particular, our work centers on a representative deployment in the city of Toronto, while remaining applicable to any global urban environment exhibiting similar co-channel interference conditions. The objective is to design and validate an intelligent spectrum and power allocation mechanism capable of effectively mitigating terrestrial co-channel interference.

A. System Architecture

We consider a DtC system consisting of one or multiple LEO satellites, which provides NR downlink services to K UEs within a coverage area $\mathcal{A} \subset \mathbb{R}^2$. The UE set is denoted as

$$\mathcal{K} = \{1, 2, \ldots, K\}, \tag{1}$$

and the ground position of UE k is represented by

$$\mathbf{u}_k = [x_k,\, y_k]^\mathsf{T} \in \mathcal{A}, \tag{2}$$

where the spatial distribution of UEs follows a homogeneous or inhomogeneous Poisson point process, so as to capture realistic user distributions in urban environments.

The system operates over a frequency band composed of Z orthogonal Physical Resource Blocks (PRBs), denoted by the index set

$$Z = \{1, \ldots, Z\}. \tag{3}$$

In DtC scenarios, the satellite downlink carrier is typically reused by terrestrial LTE/NR systems. To avoid the uncertainty and complexity of explicitly modeling each terrestrial base station, we employ an externally constructed three-dimensional electromagnetic map (Radio Map). This Radio Map is generated by importing realistic 3D city building models into Sionna-RT, performing full-path ray tracing (direct, reflected, diffracted, and scattered components), and applying low-rank reconstruction (TNNR-ADMM) to fill missing data. For each location (x, y) and PRB n, the Radio Map provides an accurate prediction of terrestrial interference power:

$$R(x, y, n) \text{ [dBm]}. \tag{4}$$

During scheduling, the terrestrial interference observed by UE k on PRB n is obtained by querying the Radio Map:

$$I_{k,n} = 10^{R(x_k, y_k, n)/10}. \tag{5}$$

This mechanism replaces traditional PRB-level interference measurements and CQI/CSI feedback, fully avoiding the CQI aging problem inherent in NTN environments.

The satellite orbit is provided by a high-fidelity orbit model driven by Two-Line Element (TLE) files. At each Transmission Time Interval (TTI), the orbit module outputs the satellite's 3D position $\mathbf{s}(t)$, the instantaneous slant range $d_k(t)$, elevation angle $e_k(t)$, Doppler shift, and propagation delay for each UE. These parameters strictly follow the geometric assumptions specified in 3GPP TR 38.811 and TR 38.821.

Based on all available information (large-scale channel conditions, beam gains, and Radio Map predictions), the centralized scheduler—located either onboard the satellite or at a gateway—makes per-TTI resource allocation decisions, including:

- PRB assignment $x_{k,n} \in \{0, 1\}$,
- PRB power allocation $p_{k,n} \geq 0$.

The overall system architecture is illustrated in **Fig. 2**. The LEO satellite communicates with UEs via a service link, while the centralized scheduler determines resource allocation within each beam footprint.

B. NTN Downlink Channel Model

This subsection establishes the physical propagation model for the direct-to-cell service link. Adhering to the 3GPP TR 38.811 specification, we integrate dynamic orbital geometry with a three-state statistical fading model to determine the received signal power.

The received power for user $k \in \mathcal{K}$ on PRB $n \in \mathcal{Z}$ at time t, denoted as $P_{\text{rx}}(k, n, t)$ (in dBm), is calculated via the following link budget:

$$\begin{aligned} P_{\text{rx,dBm}}(k, n, t) &= P_{\text{tx,dBm}}(n, t) + G_{k,\text{dBi}}(t) \\ -L_{\text{fs,dB}}(k, t) &- L_{\text{LS,dB}}(k, t) + F_{\text{small,dB}}(k, n, t), \end{aligned} \tag{6}$$

where the components are defined as follows:

Transmit Power (P_{tx}): $P_{\text{tx,dBm}}(n, t)$ represents the satellite's Equivalent Isotropically Radiated Power (EIRP) allocated to PRB n. This serves as the primary decision variable in the power domain optimization.

Free-Space Path Loss (L_{fs}): Given the time-varying slant distance $d_k(t)$ (km) between the LEO satellite and the user, the path loss is modeled as:

$$L_{\text{fs,dB}}(k, t) = 32.45 + 20 \log_{10} \big(d_k(t)\big) + 20 \log_{10}(f_c), \tag{7}$$

where f_c denotes the carrier frequency in MHz.

Satellite Beam Gain (G_k): To capture the directivity of the satellite spot beam, the antenna gain $G_{k,\text{dBi}}(t)$ is modeled as a function of the user's instantaneous off-axis angle $\theta_k(t)$ relative to the beam boresight:

$$G_{k,\text{dBi}}(t) = G_{\max} + 10 \log_{10} \left([\cos(\theta_k(t))]^m \right), \tag{8}$$

where $G_{\max}$ is the peak boresight gain, and the exponent m determines the beam roll-off factor, ensuring realistic spot beam coverage.

Large-Scale and Small-Scale Fading: The channel implements the 3GPP three-state model. The link state $S_k(t) \in \{\text{LoS}, \text{SLoS}, \text{NLoS}\}$ is determined probabilistically based on the user's elevation angle. The large-scale fading $L_{\text{LS,dB}}(k, t)$ includes state-specific excess loss and log-normal shadowing $X_{\sigma, S_k(t)}$. The small-scale fading term $F_{\text{small,dB}}(k, n, t)$ follows a Rician distribution for LoS/SLoS states and a Rayleigh distribution for NLoS states. Consistent with the system bandwidth, small-scale fading is assumed to be frequency-flat per PRB but identically distributed (i.i.d.) across the frequency domain.

C. Interference Model

The total interference-plus-noise power, $I_{\text{total},k,n}(t)$ (in mW) is defined as:

$$I_{\text{total},k,n}(t) = I_{k,n}(t) + N_0^{\text{eff}}, \tag{9}$$

where $I_{k,n}(t)$ represents the aggregate terrestrial downlink interference power obtained by querying the pre-constructed Radio Map at the user's coordinates (x_k, y_k) for PRB n. This allows the system to capture realistic, frequency-selective interference patterns caused by urban topology.

The effective noise power N_0^{eff} (in mW) accounts for thermal noise and receiver imperfections. In the logarithmic domain, it is expressed as:

$$N_{0,\text{dBm}}^{\text{eff}} = -174 + 10 \log_{10}(B_{\text{PRB}}) + F_{\text{NF,dB}} + L_{\text{impl,dB}}, \tag{10}$$

where -174 dBm/Hz is the thermal noise power spectral density at the reference temperature of 290 K, B_{PRB} is the PRB bandwidth, $F_{\text{NF,dB}}$ is the user receiver noise figure, and $L_{\text{impl,dB}}$ represents implementation losses.

D. Link Quality Abstraction and Mapping

To bridge the physical layer SINR to the MAC layer resource allocation, we employ an abstraction model that accounts for the contiguous resource allocation constraint in 5G NR.

Instantaneous SINR: The Signal-to-Interference-plus-Noise Ratio (SINR) for user k on PRB n is derived from the signal and interference models:

$$\Gamma_{k,n}(t) = \frac{10^{P_{\text{rx,dBm}}(k,n,t)/10}}{I_{\text{total},k,n}(t)}. \tag{11}$$

Effective SINR Aggregation (EESM): Since a user is allocated a contiguous block of PRBs $\mathcal{Z}_k$ employing a single Modulation and Coding Scheme (MCS), we utilize the Exponential Effective SINR Mapping (EESM) to compress the vector of PRB-level SINRs into a scalar metric:

$$\Gamma_{k,\text{eff}}(t) = -\beta \ln\left(\frac{1}{|\mathcal{Z}_k|}\sum_{n\in\mathcal{Z}_k}\exp\left(-\frac{\Gamma_{k,n}(t)}{\beta}\right)\right), \quad (12)$$

where $\mathcal{Z}_k$ represents the set of contiguous PRBs allocated to user k and β is a calibration parameter dependent on the specific MCS.

Spectral Efficiency Mapping: Finally, the achievable Spectral Efficiency (SE) is obtained by mapping the effective SINR to the discrete MCS tables specified in 3GPP TS 38.214 [8]:

$$SE_k(\mathcal{Z}_k, t) = f_{\text{MCS}}\big(\Gamma_{k,\text{eff}}(t)\big). \quad (13)$$

This mapping $f_{\text{MCS}}(\cdot)$ provides the realistic bits/s/Hz achievable for the allocated block, capturing the "cliff effect" of discrete link adaptation.

E. Problem Formulation

The primary objective is to maximize the total system spectral efficiency through the joint optimization of spectrum (PRB) allocation and power control. We consider a discrete-time resource allocation problem where, in each TTI t, the scheduler determines the optimal PRB assignment matrix $\mathbf{X}(t)$ and the power vector $\mathbf{p}(t)$.

Let $x_{k,n}(t) \in \{0,1\}$ denote the binary scheduling variable indicating whether PRB n is assigned to UE k in TTI t:

$$x_{k,n}(t) = \begin{cases} 1, & \text{if PRB } n \text{ is allocated to UE } k \text{ in TTI } t,\\ 0, & \text{otherwise.} \end{cases} \quad (14)$$

The set of PRBs allocated to UE k in TTI t is

$$\mathcal{Z}_k(t) = \{n \in \mathcal{Z} \mid x_{k,n}(t) = 1\}. \quad (15)$$

According to the NR specification, $\mathcal{Z}_k(t)$ must contain at most one contiguous PRB block; this constraint is encoded by the feasible scheduling set $\mathcal{C}$.

Let $p_n(t) \in \mathbb{R}_{\geq 0}$ denote the transmit power allocated to PRB n at TTI t. Since at most one UE may occupy PRB n in a given TTI, it is sufficient to define power at the PRB level rather than per UE–PRB pair. Note that $p_n(t)$ corresponds to the linear domain value of $P_{\text{tx,dBm}}(n,t)$ defined in the link budget.

Based on the physical-layer model in Section II-B, the instantaneous SINR of UE k on PRB n in TTI t is written explicitly as a function of the PRB power:

$$\Gamma_{k,n}(t) = \Gamma_{k,n}\big(p_n(t)\big). \quad (16)$$

Accordingly, the instantaneous SE for UE k on PRB n in TTI t becomes

$$SE_{k,n}(t) = f(\Gamma_{k,n}(t)) = f\big(\Gamma_{k,n}\big(p_n(t)\big)\big) \triangleq SE_{k,n}\big(p_n(t)\big). \quad (17)$$

Over a scheduling horizon of T TTIs, the aggregate system spectral efficiency is

$$\sum_{t=1}^{T}\sum_{k\in\mathcal{K}}\sum_{n\in\mathcal{Z}} x_{k,n}(t)\, SE_{k,n}\big(p_n(t)\big). \quad (18)$$

Define the active PRB set in TTI t as

$$\mathcal{Z}_{\text{active}}(t) = \{n \in \mathcal{Z} \mid \exists\, k \in \mathcal{K} \text{ such that } x_{k,n}(t) = 1\}. \quad (19)$$

The dynamic resource allocation problem can therefore be formulated as the following joint optimization over scheduling and power variables:

$$\begin{aligned}
\max_{\{\mathbf{X}(t),\,\mathbf{p}(t)\}_{t=1}^{T}} \quad & \sum_{t=1}^{T}\sum_{k\in\mathcal{K}}\sum_{n\in\mathcal{Z}} x_{k,n}(t)\, SE_{k,n}\big(p_n(t)\big) && (20a)\\
\text{s.t.} \quad & \sum_{k\in\mathcal{K}} x_{k,n}(t) \leq 1, \quad \forall n \in \mathcal{Z}, \forall t, && (20b)\\
& \sum_{n\in\mathcal{Z}} p_n(t) \leq P_{\text{total}}, \quad \forall t, && (20c)\\
& P_{\min} \leq p_n(t) \leq P_{\max}, \quad \forall n \in \mathcal{Z}_{\text{active}}(t), \forall t, && (20d)\\
& \mathbf{X}(t) \in \mathcal{C}, \quad \forall t, && (20e)\\
& x_{k,n}(t) \in \{0,1\}, \quad \forall k \in \mathcal{K}, \forall n \in \mathcal{Z}, \forall t. && (20f)
\end{aligned}$$

The constraints in (20b)–(20f) characterize the fundamental physical and regulatory limitations of the DtC downlink system. Specifically, constraint (20b) enforces orthogonality in the frequency domain by ensuring that each PRB is assigned to at most one UE in a given TTI, thus preventing intra-beam interference. The Constraint (20c) restricts the total transmit power across all PRB to the available power budget of the satellite, while the constraints per-PRB box in (20d) bound the individual PRB power levels. The lower bound $P_{\min}$ prevents ineffective low-power transmissions under strong terrestrial interference, whereas the upper bound $P_{\max}$ reflects regulatory spectral masks and power-amplifier linearity limits. The structural constraint (20e) requires that the scheduled PRBs for each UE form a single contiguous block, which is mandated by the NR resource-grid design and enables the use of a single MCS per allocation. Finally, (20f) captures the inherent combinatorial nature of PRB scheduling.

Complexity Analysis: The problem (20a) constitutes a non-convex Mixed-Integer Non-Linear Program (MINLP), which is known to be NP-hard. The complexity stems from the coupling of binary scheduling variables with continuous power allocation, the non-linear SINR-to-SE mapping, and the combinatorial nature of the contiguous allocation constraint. Given the millisecond-level scheduling requirements and high Doppler dynamics of LEO networks, obtaining a global optimum via exhaustive search is computationally infeasible. Consequently, we decompose the original MINLP into two tractable sub-problems: a marginal-gain-driven greedy spectrum allocation and a constrained water-filling power control, as detailed in Section III.

III. Proposed Dynamic Spectrum and Power Allocation Algorithm

To efficiently solve the joint optimization problem formulated in Section II-E, we propose a two-stage heuristic framework that decouples the discrete scheduling decisions from the continuous power control. As illustrated in the system architecture, the algorithm executes the following two sequential sub-problems at each TTI:

1) *Spectrum Allocation:* This stage determines the binary scheduling matrix $\mathbf{X}(t)$ using a marginal-gain-based greedy strategy. It explicitly exploits the heterogeneity of PRB-level interference to construct feasible, contiguous PRB blocks $\mathcal{Z}_k(t)$ for each user.
2) *Power Allocation:* Given the determined schedule, this stage optimizes the continuous transmit power vector $\mathbf{p}(t)$ for the active PRBs using a constrained water-filling approach to maximize the aggregate spectral efficiency.

The detailed procedures for these two stages are presented in the following subsections. For clarity, we omit the TTI index t in the remainder of this section unless ambiguity arises.

A. Marginal Gain-based Dynamic Spectrum Allocation Algorithm

The first stage focuses on determining the set of active PRBs and their assignment to users. We propose a **Marginal Gain-based Greedy (MGG)** algorithm that iteratively constructs contiguous resource blocks for each user. This approach leverages the Radio Map to predict interference distribution across the frequency domain, enabling the scheduler to exploit frequency diversity gain while strictly adhering to the contiguity constraint $\mathcal{C}$.

1) Metric Definition and Marginal Utility: To balance system throughput and user fairness, we adopt the Proportional Fair (PF) criterion. The scheduling decision is driven by the marginal increase in the system's total utility. Let $\bar{R}_k(t)$ denote the exponentially smoothed average historical throughput of user k at time t, updated as:

$$\bar{R}_k(t+1) = (1-\alpha)\bar{R}_k(t) + \alpha R_k(t), \tag{21}$$

where $\alpha \in (0,1)$ is the forgetting factor and $R_k(t)$ is the instantaneous data rate.

Given the contiguous block $\mathcal{Z}_k$ currently assigned to user k, the estimated instantaneous rate is calculated using EESM to account for frequency-selective interference:

$$\widehat{R}_k(\mathcal{Z}_k) = |\mathcal{Z}_k| \cdot B_{\text{PRB}} \cdot f_{\text{MCS}}(\Gamma_{k,\text{eff}}(\mathcal{Z}_k)), \tag{22}$$

where B_{PRB} is the bandwidth of a single PRB, and $f_{\text{MCS}}(\cdot)$ maps the effective SINR to the achievable spectral efficiency (bits/s/Hz) according to the 3GPP MCS tables. The effective SINR $\Gamma_{k,\text{eff}}(\mathcal{Z}_k)$ is obtained by applying EESM to the per-PRB SINRs $\{\Gamma_{k,n}\}_{n\in\mathcal{Z}_k}$, where each $\Gamma_{k,n}$ is computed from the Radio Map interference values as detailed in Section II-D.

We define the **Marginal PF Utility**, ΔU_k, as the normalized increase in throughput resulting from appending a candidate PRB set $\Delta\mathcal{Z}_k$ to the current allocation:

$$\Delta U_k(\Delta\mathcal{Z}_k) = \frac{\widehat{R}_k(\mathcal{Z}_k \cup \Delta\mathcal{Z}_k) - \widehat{R}_k(\mathcal{Z}_k)}{\bar{R}_k(t)+\epsilon}, \tag{23}$$

where ϵ is a small constant for numerical stability.

2) Algorithm Procedure: The MGG algorithm proceeds in iterations until all PRBs in the set $\mathcal{Z}$ are assigned or no further valid assignments are possible. To strictly satisfy the contiguous allocation constraint (20e) in Section II-C, the scheduler evaluates specific valid actions for each user at each step:

- **Seed Action:** If user k has no assigned PRBs ($\mathcal{Z}_k = \emptyset$), the algorithm evaluates every unallocated PRB $n \in \mathcal{Z}_{\text{free}}$ as a potential starting point ("seed"). The marginal utility here is the rate achievable on that single PRB normalized by the average throughput.
- **Grow Action:** If user k already holds a contiguous block $\mathcal{Z}_k = [n_{\text{start}}, n_{\text{end}}]$, the algorithm evaluates extending the block to its immediate neighbors. Specifically, it checks if the left neighbor $(n_{\text{start}}-1)$ or the right neighbor $(n_{\text{end}}+1)$ is available in $\mathcal{Z}_{\text{free}}$.

The detailed procedure is summarized in **Algorithm 1**. Our proposed strategy differs fundamentally from conventional wideband-CQI PF scheduling by combining fine-grained Radio Map interference awareness with block-level EESM evaluation. Specifically, by rigorously computing the marginal PF utility of each candidate expansion, the scheduler naturally avoids appending severely interfered PRBs that would reduce the effective SINR and thereby force a lower MCS across the entire block—a capability that is particularly important under pronounced PRB-to-PRB interference heterogeneity. Consequently, the greedy heuristic prioritizes PRBs with low terrestrial interference for users that can exploit them most effectively, improving the instantaneous system spectral efficiency while maintaining PRB contiguity through the restricted seed-and-grow action set.

B. Group-Based Power Allocation Optimization

Once the binary scheduling matrix $\mathbf{X}(t)$ is determined by the MGG algorithm, the original MINLP (20a) reduces to a continuous power allocation problem over the set of active PRBs.As formulated in (20c) and (20d), the objective is to optimize the transmit power vector $\mathbf{p}(t) = [p_1(t), \ldots, p_Z(t)]$ to maximize the instantaneous aggregate spectral efficiency, subject to the total power budget P_{total} and per-PRB box constraints.

1) Problem Reformulation: While classical water-filling operates on individual subchannels, the contiguous NR allocation constraint implies that a single MCS is applied across the entire block $\mathcal{Z}_k(t)$ assigned to user k. Excessive power variation among the PRBs within such a block can exacerbate the frequency-selective fading effects, leading to a penalty in the effective SINR computed via EESM. To mitigate this situation and simplify optimization, we adopt a group-level power allocation strategy, where a uniform power level $p_k(t)$ is assigned to every PRB $n \in \mathcal{Z}_k(t)$.

Let $W_k = |\mathcal{Z}_k(t)|$ denote the bandwidth (in number of PRBs) allocated to user k, and let $\mathcal{K}_{\text{active}}(t)$ be the set of scheduled users. Consistent with the group-level constraint, the transmit power is uniform across the allocated block, i.e., $p_n(t) = p_k(t)$ for all $n \in \mathcal{Z}_k(t)$.

Algorithm 1 Marginal Gain-based Greedy (MGG) Spectrum Allocation

Require: Set of users $\mathcal{K}$, set of PRBs $\mathcal{Z}$, Radio Map interference $\{I_{k,n}\}$, historical rates $\{\bar{R}_k(t)\}$
Ensure: Scheduling matrix $\mathbf{X}$
Initialize $\mathcal{Z}_{\text{free}} \leftarrow \mathcal{Z}$, $\mathcal{Z}_k \leftarrow \emptyset$ for all $k \in \mathcal{K}$, and $x_{k,n} \leftarrow 0$ for all (k,n)
while $\mathcal{Z}_{\text{free}} \neq \emptyset$ **do**
  $\Delta U_{\max} \leftarrow -\infty$, $k^* \leftarrow -1$, $n^* \leftarrow -1$
  for all $k \in \mathcal{K}$ **do**
    Determine candidate PRBs $\mathcal{N}_k$ based on contiguity:
    if $\mathcal{Z}_k = \emptyset$ **then**
      $\mathcal{N}_k \leftarrow \mathcal{Z}_{\text{free}}$ {**Seed candidates**}
      $\widehat{R}_k^{\text{current}} \leftarrow 0$
    else
      $n_{\text{start}} \leftarrow \min(\mathcal{Z}_k)$, $\quad n_{\text{end}} \leftarrow \max(\mathcal{Z}_k)$
      $\mathcal{N}_k \leftarrow \{n_{\text{start}} - 1, n_{\text{end}} + 1\} \cap \mathcal{Z}_{\text{free}}$ {**Grow candidates**}
      Calculate $\widehat{R}_k^{\text{current}} \triangleq \widehat{R}_k(\mathcal{Z}_k)$ using EESM
    end if
    for all $n \in \mathcal{N}_k$ **do**
      Calculate $\widehat{R}_k^{\text{new}} \triangleq \widehat{R}_k(\mathcal{Z}_k \cup \{n\})$ using EESM
      Define $\Delta\mathcal{Z}_{k,n} \triangleq \{n\}$ and compute $\Delta U_{k,n} \leftarrow \Delta U_k(\Delta\mathcal{Z}_{k,n})$ via (23)
      if $\Delta U_{k,n} > \Delta U_{\max}$ **then**
        $\Delta U_{\max} \leftarrow \Delta U_{k,n}$, $k^* \leftarrow k$, $n^* \leftarrow n$
      end if
    end for
  end for
  if $k^* \neq -1$ **then**
    Assign PRB n^* to user k^*: $x_{k^*,n^*} \leftarrow 1$
    Update sets: $\mathcal{Z}_{k^*} \leftarrow \mathcal{Z}_{k^*} \cup \{n^*\}$, $\mathcal{Z}_{\text{free}} \leftarrow \mathcal{Z}_{\text{free}} \setminus \{n^*\}$
  else
    break {No valid assignments possible}
  end if
end while
$\mathbf{X} \leftarrow [x_{k,n}]_{k\in\mathcal{K},\, n\in\mathcal{Z}}$
return $\mathbf{X}$

Although the actual system throughput is determined by the discrete MCS mapping function $f(\cdot)$ defined in (17), this stepwise function is non-differentiable and non-convex, rendering standard gradient-based optimization intractable. To address this, we approximate $f(\cdot)$ using the Shannon capacity formula. This approach serves as a smooth, differentiable proxy that captures the monotonic relationship between SINR and spectral efficiency, thereby transforming the power allocation sub-problem into a tractable convex optimization formulation:

$$\max_{\{p_k(t)\}_{k\in\mathcal{K}_{\text{active}}}} \quad \sum_{k\in\mathcal{K}_{\text{active}}} W_k \log_2\left(1 + p_k(t)\cdot \bar{H}_k(t)\right) \tag{24a}$$
$$\text{s.t.} \quad \sum_{k\in\mathcal{K}_{\text{active}}} W_k\, p_k(t) \le P_{\text{total}}, \tag{24b}$$
$$P_{\min} \le p_k(t) \le P_{\max}, \quad \forall k \in \mathcal{K}_{\text{active}}. \tag{24c}$$

Here, $\bar{H}_k(t)$ represents the effective channel-to-interference-plus-noise ratio (CINR) gain for the resource block $\mathcal{Z}_k(t)$, defined as the arithmetic mean of the per-PRB channel qualities:

$$\bar{H}_k(t) = \frac{1}{W_k} \sum_{n\in\mathcal{Z}_k(t)} \frac{g_{k,n}(t)}{I_{k,n}(t) + N_0^{\text{eff}}}, \tag{25}$$

where $g_{k,n}(t)$ denotes the aggregate channel power gain in linear scale. This term encapsulates all propagation effects derived in Section II-B, including the free-space path loss, the receive antenna gain ($G_{k,\text{dBi}}$), and both large-scale and small-scale fading components:

$$g_{k,n}(t) = 10^{\frac{G_{k,\text{dBi}}(t) - L_{\text{fs,dB}}(k,t) - L_{\text{LS,dB}}(k,t) + F_{\text{small,dB}}(k,n,t)}{10}}. \tag{26}$$

2) Structure of the Optimal Power Allocation: Since problem (24) is convex, its optimal solution satisfies the Karush-Kuhn-Tucker (KKT) conditions. By formulating the Lagrangian, we derive the optimal power allocation structure as a water-filling solution with box constraints:

$$p_k^*(t) = \left[\frac{1}{\nu} - \frac{1}{\bar{H}_k(t)}\right]_{P_{\min}}^{P_{\max}}, \tag{27}$$

where $[x]_a^b = \min(\max(x,a), b)$ denotes the projection onto the interval $[a,b]$, and $\nu > 0$ is the Lagrange multiplier (inverse water level) associated with the total power constraint (24b). The value of ν is chosen such that the power budget equality holds:

$$\sum_{k\in\mathcal{K}_{\text{active}}} W_k\, p_k^*(t) \approx P_{\text{total}}. \tag{28}$$

3) Iterative Algorithm Design : Since $p_k^*(\nu)$ is a monotonically decreasing function of ν, we employ an efficient iterative bisection search to find the unique optimal water level ν^*. The algorithm iteratively adjusts ν, clamps the resulting power values to $[P_{\min}, P_{\max}]$, and checks the aggregate power consumption against P_{total}.

A feasibility condition must be addressed: if $\sum_k W_k P_{\min} > P_{\text{total}}$, the problem is infeasible under the current schedule and the per-block minimum power constraint cannot be satisfied for all scheduled users. In such cases, we deactivate the scheduled block with the smallest effective gain $\bar{H}_k(t)$ (i.e., the block contributing least to the objective) and repeat the feasibility check until the remaining set becomes feasible. The complete procedure is detailed in **Algorithm 2**.

Finally, after obtaining $\{p_k^*(t)\}$, the PRB-level power vector is recovered by setting $p_n^*(t) = p_k^*(t)$ for all $n \in \mathcal{Z}_k(t)$, and $p_n^*(t) = 0$ for unallocated PRBs. The realized throughput for performance evaluation can then be computed using the original EESM-based effective SINR and the discrete 3GPP MCS mapping.

This two-stage framework effectively decouples the discrete scheduling decisions from the continuous power optimization. By utilizing the Radio Map to guide the greedy allocation away from high-interference bands in the first stage, and then applying water-filling to maximize the rate on the selected resources in the second stage, the proposed algorithm achieves a robust approximation of the joint optimum with low computational complexity suitable for real-time satellite onboard processing.

Algorithm 2 Constrained Water-Filling Power Optimization

Require: Scheduled blocks $\{\mathcal{Z}_k(t)\}_{k\in\mathcal{K}_{\text{active}}}$, effective gains $\{\bar{H}_k(t)\}$, constraints $P_{\text{total}}, P_{\min}, P_{\max}$.
Ensure: Optimal power vector $\mathbf{p}^*(t) = \{p_n^*(t)\}_{n\in\mathcal{Z}}$.
Calculate block widths $W_k \leftarrow |\mathcal{Z}_k(t)|$ for all active k.
Feasibility Check:
while $\sum_k W_k P_{\min} > P_{\text{total}}$ **do**
  Find user with worst channel: $k_{\text{worst}} \leftarrow \arg\min_k \bar{H}_k(t)$.
  Drop user: $\mathcal{K}_{\text{active}} \leftarrow \mathcal{K}_{\text{active}} \setminus \{k_{\text{worst}}\}$.
end while
Initialize water level search interval $[\nu_{\min}, \nu_{\max}]$.
while $|\nu_{\max} - \nu_{\min}| > \delta$ **do**
  $\nu \leftarrow (\nu_{\min} + \nu_{\max})/2$.
  for all $k \in \mathcal{K}_{\text{active}}$ **do**
    Tentative power: $\tilde{p}_k \leftarrow \frac{1}{\nu} - \frac{1}{\bar{H}_k(t)}$.
    Apply box constraints: $p_k \leftarrow [\tilde{p}_k]_{P_{\min}}^{P_{\max}}$
  end for
  Total power used: $P_{\text{used}} \leftarrow \sum_k W_k\, p_k$.
  if $P_{\text{used}} > P_{\text{total}}$ **then**
    $\nu_{\min} \leftarrow \nu$ {Over budget: increase ν to reduce power}
  else
    $\nu_{\max} \leftarrow \nu$ {Under budget: decrease ν to increase power}
  end if
end while
Final Assignment:
for all $k \in \mathcal{K}_{\text{active}}$ and $n \in \mathcal{Z}_k(t)$ **do**
  $p_n^*(t) \leftarrow p_k$ {Uniform power within block}
end for
Set $p_n^*(t) \leftarrow 0$ for all unallocated PRBs.
return $\mathbf{p}^*(t)$.

IV. Performance Evaluation

A. Simulation Setup

To rigorously evaluate the performance of our proposed algorithm, we developed a system-level simulator compliant with the 3GPP NTN specifications (TR 38.811 and TR 38.821). The simulation framework effectively integrates realistic orbital dynamics, 3D ray-tracing-based interference maps, and full-stack PHY/MAC layer procedures.

We consider an S-band DtC downlink scenario operating at a carrier frequency of 2.19 GHz. The system bandwidth is configured to approximately 20 MHz, consisting of $Z = 51$ Physical Resource Blocks (PRBs) with a subcarrier spacing (SCS) of 30 kHz. This numerology is consistent with 5G NR FR1 specifications.

The satellite orbit follows the LEO parameters of the Starlink constellation (shell altitude ≈ 550 km). The orbital motion is modeled using high-precision TLE sets, updated dynamically with a temporal resolution of 1 ms per TTI. On the user side, we deploy $K = 100$ DtC-capable UEs distributed within the coverage area.

The channel model adopts the 3GPP TR 38.811 Large Scale Fading (LSF) model for S-band handheld terminals in an urban environment. It incorporates elevation-dependent probabilities for LoS, SLoS, and NLoS states, along with state-specific log-normal shadowing and Rician/Rayleigh small-scale fading. The key simulation parameters are summarized in **Table I**.

TABLE I
SIMULATION PARAMETERS

Parameter	Value
Carrier Frequency (f_c)	2.19 GHz
System Bandwidth	20 MHz (51 PRBs × 360 kHz)
Subcarrier Spacing	30 kHz
Simulation Duration (T)	2000 TTIs (2 s)
Number of UEs (K)	100
Satellite Altitude	≈ 550 km (Starlink TLE)
Max Satellite Antenna Gain	38 dBi
UE Noise Figure	7 dB
Implementation Loss	1 dB
Total Satellite EIRP Budget (P_{total})	50 dBm (Dynamic Sharing)
Baseline CSI Feedback Delay	12 TTIs
CSI Reporting Periodicity	5 TTIs

Unlike stochastic interference models, we utilize a site-specific Radio Map constructed via high-fidelity ray tracing. We imported the 3D building geometry of Toronto City into the Sionna-RT simulation platform to compute path loss and propagation effects from terrestrial base stations. The resulting map provides instantaneous terrestrial interference power $I_{k,n}$ for each PRB and geographical location, serving as the ground truth for interference awareness.

B. Baseline Scheme

To benchmark the proposed Radio Map-aware dynamic spectrum sharing (RM-DSS) algorithm, we implement a standard 3GPP-compliant baseline scheme. The baseline employs a PF scheduler that relies on conventional CSI feedback from UEs.

Power allocation for the baseline adopts a water-filling strategy constrained by the same per-PRB and total power limits as the proposed method, ensuring a fair comparison of resource management gain rather than power gain.

C. Test Scenarios

To comprehensively assess the proposed algorithm's efficacy in mitigating real-world urban interference, we define two complementary test scenarios centered on the Toronto metropolitan area.

1) Scenario A: Toronto Single-Satellite : This scenario focuses on a single satellite pass event to isolate the performance of the resource allocation algorithm from handover dynamics.

- **Orbit**: We utilize the TLE data of a specific Starlink satellite (*STARLINK-11090*) during a visible overpass above Toronto (43.69°N, 79.37°W). The simulation starts at a specific epoch (2025-10-07T21:38 UTC), capturing the dynamic variation of elevation angles and Doppler shifts.
- **Interference Environment**: A high-resolution Radio Map (approx. 125 m grid size) of Toronto is used. Users are randomly distributed across the city center, experiencing heterogeneous interference levels ranging from deep terrestrial blocking to strong Line-of-Sight interference.

- **Objective**: To quantify the instantaneous spectral efficiency gain achieved by replacing delayed feedback with real-time Radio Map awareness under specific geometric conditions.

2) Scenario B: Toronto Constellation: This scenario extends the evaluation to a continuous service model involving a mega-constellation.

- **Constellation Configuration**: A complete shell of Starlink satellites is loaded from the TLE catalog, simulating a realistic multi-satellite environment.
- **Dynamic Handover**: The simulation enables dynamic satellite selection and handover. A user associates with the satellite that maximizes the reference signal received power (RSRP) or wideband SINR. A handover hysteresis margin of 2.0 dB and a Time-to-Trigger (TTT) of 20 TTIs are applied to prevent ping-pong effects.
- **Objective**: To verify the robustness of the proposed algorithm in a complex network environment where satellite switching and beam handovers occur frequently, assessing its long-term stability and consistency.

D. Simulation Results

1) Analysis of Spectral Efficiency Gains: Fig. 2 and Fig. 3 illustrate the spectral efficiency (SE) performance of the proposed RM-DSS algorithm compared to the 3GPP baseline across the two defined scenarios. The results provide compelling evidence of the advantages of utilizing environment-aware radio maps over legacy feedback mechanisms in high-latency NTN links.

a) Scenario A: Toronto Single-Satellite: As shown in Fig. 2, in the single-satellite overpass scenario, the baseline scheme achieves an average SE of 0.116 bits/s/Hz. This relatively low value is attributed to the severe terrestrial interference in Toronto and the inability of the baseline scheduler to react promptly to channel notches due to the 12 ms feedback delay. In contrast, the proposed RM-DSS improves the SE to 0.148 bits/s/Hz.

Fig. 3 further quantifies this improvement, showing a relative gain of 27.26%. This performance boost stems from the Radio Map's ability to predict interference "hotspots" geographically. By preemptively avoiding PRBs with strong terrestrial interference—even before the UE reports a degraded CQI—the RM-DSS effectively converts spectrum that would have resulted in packet errors (Block Error Rate) for the baseline into useful throughput.

b) Scenario B: Toronto Constellation: In the constellation scenario, the absolute performance of both schemes increases significantly (0.314 bits/s/Hz for Baseline and 0.422 bits/s/Hz for RM-DSS). This overall uplift is expected, as the constellation availability allows UEs to dynamically hand over to satellites with higher elevation angles and better link budgets, rather than being constrained to a single, potentially low-elevation satellite.

Notably, the relative gain of the proposed RM-DSS over the baseline expands to 34.47%. This increased margin highlights the robustness of the Radio Map approach in highly dynamic environments. In a constellation setting, frequent cell switching and rapid variations in Doppler and geometry exacerbate the "aging" problem of conventional CSI feedback. The baseline struggles to stabilize modulation and coding schemes (MCS) during handover transition periods. Conversely, the RM-DSS relies on location-based priors which remain valid regardless of the serving satellite ID, ensuring consistent interference evasion and more aggressive MCS selection even during handover events.

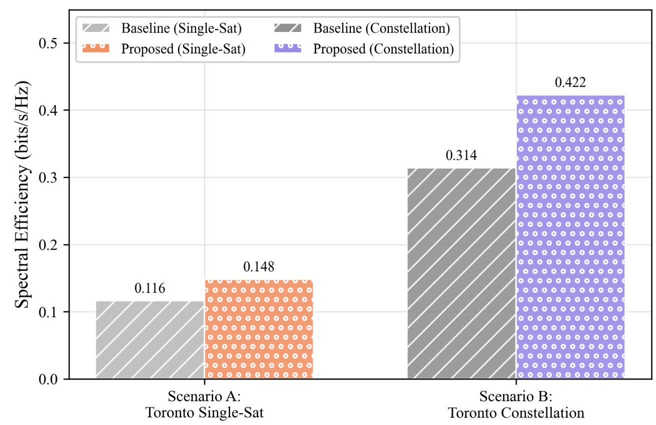

Fig. 2. Average Spectral Efficiency comparison between the 3GPP Baseline and proposed RM-DSS in Single-Satellite and Constellation scenarios.

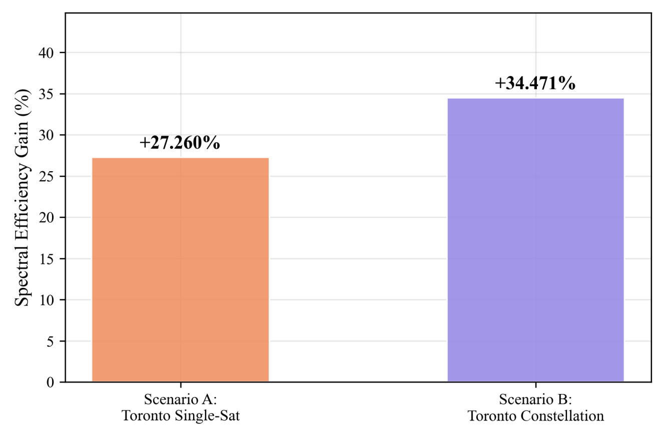

Fig. 3. Percentage Gain of Spectral Efficiency achievable by RM-DSS over the Baseline.

2) Analysis of User Throughput Performance:

3) Evaluation of HARQ Performance:

4) Robustness Analysis:

V. Relative Work

VI. Conclusion

References

[1] P. Yang, Y. Xiao, M. Xiao, and S. Li, "6g wireless communications: Vision and potential techniques," *IEEE Network*, vol. 33, no. 4, pp. 70–75, 2019.

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[3] A. Sattarzadeh, Y. Liu, A. Mohamed, R. Song, P. Xiao, Z. Song, H. Zhang, R. Tafazolli, and C. Niu, "Satellite-based non-terrestrial networks in 5g: Insights and challenges," *IEEE Access*, vol. 10, pp. 11 274–11 283, 2021.

[4] N. Heydarishahreza, T. Han, and N. Ansari, "Spectrum sharing and interference management for 6g leo satellite-terrestrial network integration," *IEEE Communications Surveys & Tutorials*, 2024.
[5] 3GPP, "Solutions for NR to Support Non-Terrestrial Networks (NTN)," 3rd Generation Partnership Project (3GPP), TR 38.821, Mar. 2023, release 16.

VII. The Design, Intent, and Limitations of the Templates

The templates are intended to **approximate the final look and page length of the articles/papers**. **They are NOT intended to be the final produced work that is displayed in print or on IEEEXplore®**. They will help to give the authors an approximation of the number of pages that will be in the final version. The structure of the LaTeX files, as designed, enable easy conversion to XML for the composition systems used by the IEEE. The XML files are used to produce the final print/IEEEXplore pdf and then converted to HTML for IEEEXplore.

VIII. Where to Get LaTeX Help — User Groups

The following online groups are helpful to beginning and experienced LaTeX users. A search through their archives can provide many answers to common questions.

http://www.latex-community.org/
https://tex.stackexchange.com/

IX. Other Resources

See [**?**], [**?**], [**?**], [**?**], [**?**] for resources on formatting math into text and additional help in working with LaTeX.

X. Text

For some of the remainer of this sample we will use dummy text to fill out paragraphs rather than use live text that may violate a copyright.

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$$x = \sum_{i=0}^{n} 2iQ. \tag{29}$$

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XI. Some Common Elements

A. Sections and Subsections

Enumeration of section headings is desirable, but not required. When numbered, please be consistent throughout the article, that is, all headings and all levels of section headings in the article should be enumerated. Primary headings are designated with Roman numerals, secondary with capital letters, tertiary with Arabic numbers; and quaternary with lowercase letters. Reference and Acknowledgment headings are unlike all other section headings in text. They are never enumerated. They are simply primary headings without labels, regardless of whether the other headings in the article are enumerated.

B. Citations to the Bibliography

The coding for the citations is made with the LaTeX `\cite` command. This will display as: see [**?**].

For multiple citations code as follows: `\cite{ref1,ref2,ref3}` which will produce [**?**], [**?**], [**?**]. For reference ranges that are not consecutive code as `\cite{ref1,ref2,ref3,ref9}` which will produce [**?**], [**?**], [**?**], [**?**]

Fig. 4. Simulation results for the network.

C. Lists

In this section, we will consider three types of lists: simple unnumbered, numbered, and bulleted. There have been many options added to IEEEtran to enhance the creation of lists. If your lists are more complex than those shown below, please refer to the original "IEEEtran_HOWTO.pdf" for additional options.

A plain unnumbered list:

bare_jrnl.tex
bare_conf.tex
bare_jrnl_compsoc.tex
bare_conf_compsoc.tex
bare_jrnl_comsoc.tex

A simple numbered list:

1) bare_jrnl.tex
2) bare_conf.tex
3) bare_jrnl_compsoc.tex
4) bare_conf_compsoc.tex
5) bare_jrnl_comsoc.tex

A simple bulleted list:

- bare_jrnl.tex
- bare_conf.tex
- bare_jrnl_compsoc.tex
- bare_conf_compsoc.tex
- bare_jrnl_comsoc.tex

D. Figures

Fig. 1 is an example of a floating figure using the graphicx package. Note that `\label` must occur AFTER (or within) `\caption`. For figures, `\caption` should occur after the `\includegraphics`.

Fig. 2(a) and 2(b) is an example of a double column floating figure using two subfigures. (The subfig.sty package must be loaded for this to work.) The subfigure `\label` commands are set within each subfloat command, and the `\label` for the overall figure must come after `\caption`. `\hfil` is used as a separator to get equal spacing. The combined width of all the parts of the figure should do not exceed the text width or a line break will occur.

Note that often IEEE papers with multi-part figures do not place the labels within the image itself (using the optional argument to `\subfloat`[]), but instead will reference/describe all of them (a), (b), etc., within the main caption. Be aware that for subfig.sty to generate the (a), (b), etc., subfigure labels, the optional argument to `\subfloat` must be present. If a subcaption is not desired, leave its contents blank, e.g.,`\subfloat`[].

TABLE II
AN EXAMPLE OF A TABLE

One	Two
Three	Four

XII. TABLES

Note that, for IEEE-style tables, the `\caption` command should come BEFORE the table. Table captions use title case. Articles (a, an, the), coordinating conjunctions (and, but, for, or, nor), and most short prepositions are lowercase unless they are the first or last word. Table text will default to `\footnotesize` as the IEEE normally uses this smaller font for tables. The `\label` must come after `\caption` as always.

XIII. ALGORITHMS

Algorithms should be numbered and include a short title. They are set off from the text with rules above and below the title and after the last line.

Algorithm 3 Weighted Tanimoto ELM.

TRAIN($\mathbf{XT}$)
  select randomly $W \subset \mathbf{X}$
  $N_{\mathbf{t}} \leftarrow |\{i : \mathbf{t}_i = \mathbf{t}\}|$ **for** $\mathbf{t} = -1, +1$
  $B_i \leftarrow \sqrt{\textsc{max}(N_{-1}, N_{+1})/N_{\mathbf{t}_i}}$ **for** $i = 1, ..., N$
  $\hat{\mathbf{H}} \leftarrow B \cdot (\mathbf{X}^T\mathbf{W})/(\nVdash\mathbf{X} + \nVdash\mathbf{W} - \mathbf{X}^T\mathbf{W})$
  $\beta \leftarrow \left(I/C + \hat{\mathbf{H}}^T\hat{\mathbf{H}}\right)^{-1} (\hat{\mathbf{H}}^T B \cdot \mathbf{T})$
  return $\mathbf{W}, \beta$

PREDICT($\mathbf{X}$)
  $\mathbf{H} \leftarrow (\mathbf{X}^T\mathbf{W})/(\nVdash\mathbf{X} + \nVdash\mathbf{W} - \mathbf{X}^T\mathbf{W})$
  return $\textsc{sign}(\mathbf{H}\beta)$

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XIV. MATHEMATICAL TYPOGRAPHY AND WHY IT MATTERS

Typographical conventions for mathematical formulas have been developed to **provide uniformity and clarity of presentation across mathematical texts**. This enables the readers

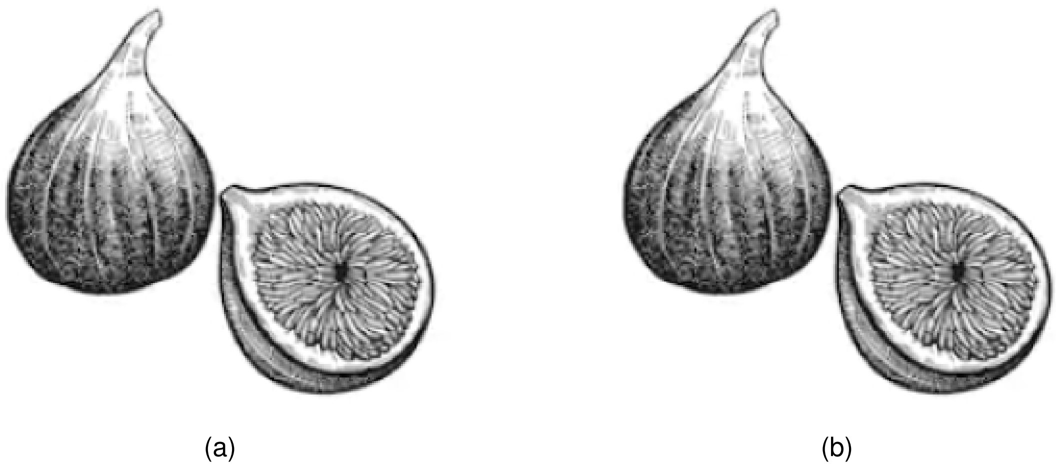

Fig. 5. Dae. Ad quatur autat ut porepel itemoles dolor autem fuga. Bus quia con nessunti as remo di quatus non perum que nimus. (a) Case I. (b) Case II.

of those texts to both understand the author's ideas and to grasp new concepts quickly. While software such as LATEX and MathType® can produce aesthetically pleasing math when used properly, it is also very easy to misuse the software, potentially resulting in incorrect math display.

IEEE aims to provide authors with the proper guidance on mathematical typesetting style and assist them in writing the best possible article. As such, IEEE has assembled a set of examples of good and bad mathematical typesetting [?], [?], [?], [?], [?].

Further examples can be found at http://journals.ieeeauthorcenter.ieee.org/wp-content/uploads/sites/7/IEEE-Math-Typesetting-Guide-for-LaTeX-Users.pdf

A. Display Equations

The simple display equation example shown below uses the "equation" environment. To number the equations, use the `\label` macro to create an identifier for the equation. LaTeX will automatically number the equation for you.

$$x = \sum_{i=0}^{n} 2iQ. \tag{30}$$

is coded as follows:

```
\begin{equation}
\label{deqn_ex1}
x = \sum_{i=0}^{n} 2{i} Q.
\end{equation}
```

To reference this equation in the text use the `\ref` macro. Please see (30)
is coded as follows:

```
Please see (\ref{deqn_ex1})
```

B. Equation Numbering

Consecutive Numbering: Equations within an article are numbered consecutively from the beginning of the article to the end, i.e., (1), (2), (3), (4), (5), etc. Do not use roman numerals or section numbers for equation numbering.
Appendix Equations: The continuation of consecutively numbered equations is best in the Appendix, but numbering as (A1), (A2), etc., is permissible.

Hyphens and Periods: Hyphens and periods should not be used in equation numbers, i.e., use (1a) rather than (1-a) and (2a) rather than (2.a) for subequations. This should be consistent throughout the article.

C. Multi-Line Equations and Alignment

Here we show several examples of multi-line equations and proper alignments.
A single equation that must break over multiple lines due to length with no specific alignment.

$$\begin{multline}\text{The first line of this example}\\ \text{The second line of this example}\\ \text{The third line of this example} \tag{31}\end{multline}$$

is coded as:

```
\begin{multline}
\text{The first line of this example}\\
\text{The second line of this example}\\
\text{The third line of this example}
\end{multline}
```

A single equation with multiple lines aligned at the = signs

$$\begin{align} a &= c+d \tag{32}\\ b &= e+f \tag{33}\end{align}$$

is coded as:

```
\begin{align}
a &= c+d \\
b &= e+f
\end{align}
```

The align environment can align on multiple points as shown in the following example:

$$x = y \qquad X = Y \qquad a = bc \tag{34}$$
$$x' = y' \qquad X' = Y' \qquad a' = bz \tag{35}$$

is coded as:

```
\begin{align}
x &= y & X & =Y & a &=bc\\
x' &= y' & X' &=Y' &a' &=bz
\end{align}
```

D. Subnumbering

The amsmath package provides a subequations environment to facilitate subnumbering. An example:

$$f = g \tag{36a}$$
$$f' = g' \tag{36b}$$
$$\mathcal{L}f = \mathcal{L}g \tag{36c}$$

is coded as:

```
\begin{subequations}\label{eq:2}
\begin{align}
f&=g \label{eq:2A}\\
f' &=g' \label{eq:2B}\\
\mathcal{L}f &= \mathcal{L}g \label{eq:2c}
\end{align}
\end{subequations}
```

E. Matrices

There are several useful matrix environments that can save you some keystrokes. See the example coding below and the output.

A simple matrix:

$$\begin{matrix} 0 & 1 \\ 1 & 0 \end{matrix} \tag{37}$$

is coded as:

```
\begin{equation}
\begin{matrix}  0 &  1 \\
1 &  0 \end{matrix}
\end{equation}
```

A matrix with parenthesis

$$\begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \tag{38}$$

is coded as:

```
\begin{equation}
\begin{pmatrix} 0 & -i \\
 i &  0 \end{pmatrix}
\end{equation}
```

A matrix with square brackets

$$\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \tag{39}$$

is coded as:

```
\begin{equation}
\begin{bmatrix} 0 & -1 \\
1 &  0 \end{bmatrix}
\end{equation}
```

A matrix with curly braces

$$\begin{Bmatrix} 1 & 0 \\ 0 & -1 \end{Bmatrix} \tag{40}$$

is coded as:

```
\begin{equation}
\begin{Bmatrix} 1 &  0 \\
0 & -1 \end{Bmatrix}
\end{equation}
```

A matrix with single verticals

$$\begin{vmatrix} a & b \\ c & d \end{vmatrix} \tag{41}$$

is coded as:

```
\begin{equation}
\begin{vmatrix} a &  b \\
c &  d \end{vmatrix}
\end{equation}
```

A matrix with double verticals

$$\begin{Vmatrix} i & 0 \\ 0 & -i \end{Vmatrix} \tag{42}$$

is coded as:

```
\begin{equation}
\begin{Vmatrix} i &  0 \\
0 & -i \end{Vmatrix}
\end{equation}
```

F. Arrays

The array environment allows you some options for matrix-like equations. You will have to manually key the fences, but there are other options for alignment of the columns and for setting horizontal and vertical rules. The argument to array controls alignment and placement of vertical rules.

A simple array

$$\left(\begin{array}{cccc} a+b+c & uv & x-y & 27 \\ a+b & u+v & z & 134 \end{array} \right) \tag{43}$$

is coded as:

```
\begin{equation}
\left(
\begin{array}{cccc}
a+b+c & uv & x-y & 27\\
a+b & u+v & z & 134
\end{array} \right)
\end{equation}
```

A slight variation on this to better align the numbers in the last column

$$\left(\begin{array}{cccr} a+b+c & uv & x-y & 27 \\ a+b & u+v & z & 134 \end{array} \right) \tag{44}$$

is coded as:

```
\begin{equation}
\left(
\begin{array}{cccr}
a+b+c & uv & x-y & 27\\
a+b & u+v & z & 134
\end{array} \right)
\end{equation}
```

An array with vertical and horizontal rules

$$\left(\begin{array}{c|c|c|r} a+b+c & uv & x-y & 27 \\ \hline a+b & u+v & z & 134 \end{array}\right) \tag{45}$$

is coded as:

```
\begin{equation}
\left(
\begin{array}{c|c|c|r}
a+b+c & uv & x-y & 27\\
a+b & u+v & z & 134
\end{array} \right)
\end{equation}
```

Note the argument now has the pipe "|" included to indicate the placement of the vertical rules.

G. Cases Structures

Many times cases can be miscoded using the wrong environment, i.e., `array`. Using the `cases` environment will save keystrokes (from not having to type the `\left\lbrace`) and automatically provide the correct column alignment.

$$z_m(t) = \begin{cases} 1, & \text{if } \beta_m(t) \\ 0, & \text{otherwise.} \end{cases}$$

is coded as follows:

```
\begin{equation*}
{z_m(t)} =
\begin{cases}
1,&{\text{if}}\ {\beta }_m(t),\\
{0,}&{\text{otherwise.}}
\end{cases}
\end{equation*}
```

Note that the "&" is used to mark the tabular alignment. This is important to get proper column alignment. Do not use `\quad` or other fixed spaces to try and align the columns. Also, note the use of the `\text` macro for text elements such as "if" and "otherwise."

H. Function Formatting in Equations

Often, there is an easy way to properly format most common functions. Use of the \ in front of the function name will in most cases, provide the correct formatting. When this does not work, the following example provides a solution using the `\text` macro:

$$d_{R}^{KM} = \underset{d_{l}^{KM}}{\text{arg min}} \{ d_{1}^{KM}, \ldots, d_{6}^{KM}\}.$$

is coded as follows:

```
\begin{equation*}
 d_{R}^{KM} = \underset {d_{l}^{KM}}
 {\text{arg min}} \{ d_{1}^{KM},
 \ldots,d_{6}^{KM}\}.
\end{equation*}
```

I. Text Acronyms Inside Equations

This example shows where the acronym "MSE" is coded using `\text{}` to match how it appears in the text.

$$\text{MSE} = \frac{1}{n}\sum_{i=1}^{n}(Y_{i} - \hat{Y_{i}})^{2}$$

```
\begin{equation*}
 \text{MSE} = \frac {1}{n}\sum _{i=1}^{n}
(Y_{i} - \hat {Y_{i}})^{2}
\end{equation*}
```

XV. Conclusion

The conclusion goes here.

Acknowledgments

This should be a simple paragraph before the References to thank those individuals and institutions who have supported your work on this article.

Appendix
Proof of the Zonklar Equations

Use `\appendix` if you have a single appendix: Do not use `\section` anymore after `\appendix`, only `\section*`. If you have multiple appendixes use `\appendices` then use `\section` to start each appendix. You must declare a `\section` before using any `\subsection` or using `\label` (`\appendices` by itself starts a section numbered zero.)

References Section

You can use a bibliography generated by BibTeX as a .bbl file. BibTeX documentation can be easily obtained at: http://mirror.ctan.org/biblio/bibtex/contrib/doc/ The IEEEtran BibTeX style support page is: http://www.michaelshell.org/tex/ieeetran/bibtex/

Simple References

You can manually copy in the resultant .bbl file and set second argument of `\begin` to the number of references (used to reserve space for the reference number labels box).

Biography Section

If you have an EPS/PDF photo (graphicx package needed), extra braces are needed around the contents of the optional argument to biography to prevent the LaTeX parser from getting confused when it sees the complicated `\includegraphics` command within an optional argument. (You can create your own custom macro containing the `\includegraphics` command to make things simpler here.)

If you include a photo:

Michael Shell Use `\begin{IEEEbiography}` and then for the 1st argument use `\includegraphics` to declare and link the author photo. Use the author name as the 3rd argument followed by the biography text.

If you will not include a photo:

John Doe Use `\begin{IEEEbiographynophoto}` and the author name as the argument followed by the biography text.